

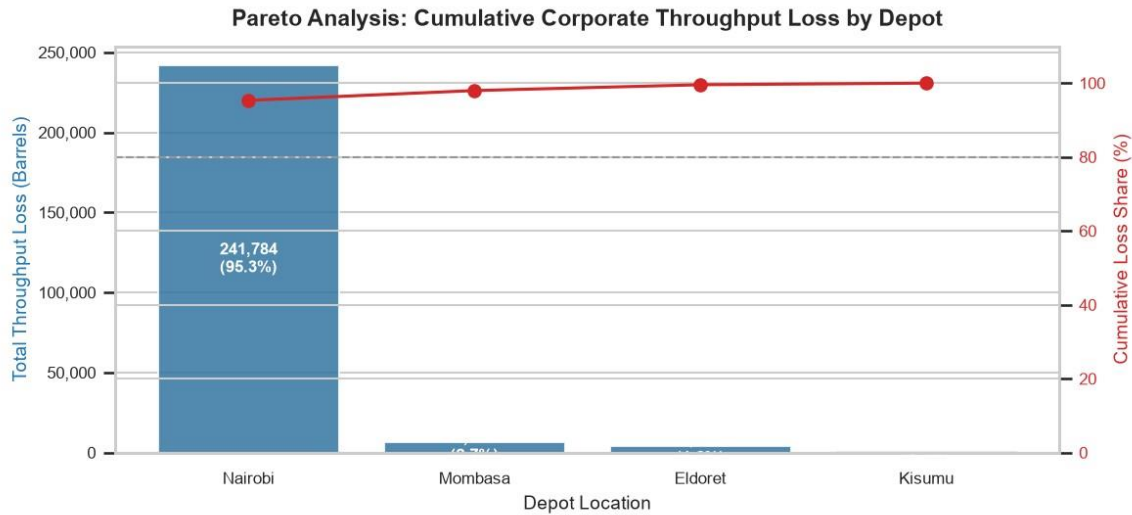
EXECUTIVE OPERATIONAL DIAGNOSTIC REPORT

Subject: Regional Depot Throughput Diagnostics & Nairobi Root Cause Analysis
Prepared for: Operations Director | Prepared by: Macharia wa Kariuki (Lead Business Detective)
Date: July 24, 2026 | Dataset Period: Full Year 2026 (2,920 records)

1. The Operational Problem (Context)

A sudden, severe decline in corporate throughput was detected starting on February 1, 2026. A review of the complete dataset of 2,920 balanced shift-depot records was conducted to verify data quality and locate the loss. The dataset was found to be highly healthy: there are 0 duplicate records, and minor missing values (operator IDs, temperatures, and throughput) were handled systematically to ensure zero analytical bias.

A comparison of shift throughput before and after February 1st reveals that the operational crisis is highly localized. While three regional depots (Eldoret, Kisumu, and Mombasa) maintained stable outputs, the Nairobi depot experienced a massive structural throughput drop of 34.4%, falling from an average of 1,052.52 barrels to 690.56 barrels per shift. Our Pareto analysis reveals that Nairobi alone accounts for 95.33% of total corporate throughput loss (representing 241,789 barrels lost in Nairobi out of 253,632 barrels lost corporate-wide).



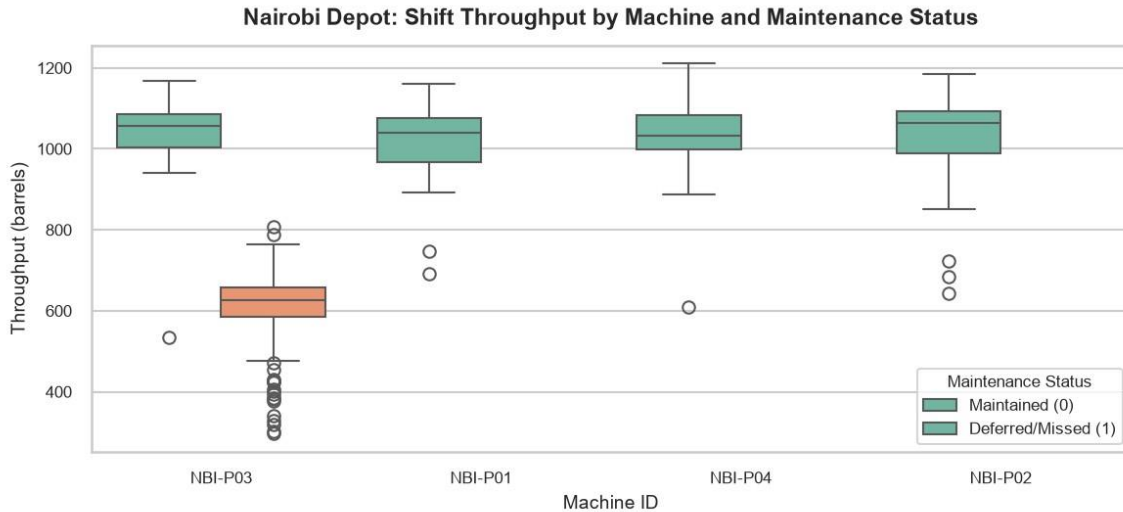
2. Root Cause Identification (Insight)

Drilling down into Nairobi's operations reveals the exact root cause of the drop: a major failure in machine scheduling and deferred maintenance on machine NBI-P03. Our analysis discovered that:

- **Machine Overuse Shift:** Prior to February 1st, Nairobi rotated its 4 machines (NBI-P01, P02, P03, P04) evenly. After February 1st, depot management shifted to running machine NBI-P03 almost exclusively-on 81.6% of all shifts (545 out of 668 shifts).
- **Missed Maintenance:** Simultaneously, `NBI-P03` was run with a pending maintenance flag (`maintenance_flag = 1`) continuously for all 545 shifts. This resulted in 529 recorded `missed_maintenance` incident events.

- **Dramatic Efficiency Loss:** When properly maintained (in January), `NBI-P03` runs at a highly efficient mean of 1,027.69 barrels per shift. Running it under deferred maintenance dropped its average output to 617.91 barrels per shift- a massive loss of 409.78 barrels per shift. This single maintenance failure explains 88.0% of the entire company's throughput loss.

- **Alternative Explanations Ruled Out:** Temperature ($r = 0.168$) and voltage ($r = -0.017$) show no correlation with throughput, ruling out weather or power grid issues. Furthermore, all 5 Nairobi operators and both shifts (Day vs. Night) saw identical throughput drops when running `NBI-P03`, ruling out labor performance issues.



3. Actionable Recommendations (Action)

To immediately recover lost throughput and prevent future operational oversights, we recommend the following directives:

1. **Immediate Maintenance Shutdown:** Take machine `NBI-P03` offline immediately for a full maintenance cycle. Restoring the machine to its maintained baseline of ~1,027 barrels will immediately recover ~410 barrels per shift (a +66% increase in its current output).
2. **Enforce Balanced Machine Rotation:** Mandate that Nairobi return to the balanced rotation policy used in January (rotating shifts evenly across NBI-P01, NBI-P02, and NBI-P04 while NBI-P03 is serviced). Running these three fully maintained machines will immediately restore Nairobi's throughput to >1,020 barrels per shift.
3. **Implement Automated Software Interlocks:** Update the central SCADA system to automatically restrict any machine from running more than 10 consecutive shifts with an active maintenance flag. If a maintenance flag is triggered, the scheduler should block the machine and force rotation to serviced backups.

SUMMARY STATS: Nairobi Post-Feb 1 Throughput Drop: -34.4% | NBI-P03 Loss due to Deferred Maintenance: -409.78 bbls/shift | Nairobi Loss Share of Corporate Total: 95.33%